SmartBand rev A schematic.
Activity band: steps, periodic heart rate, OLED, one button, haptic tap, BLE.
Global labels carry I2C, QSPI, interrupts, and the power rails between these sheets.
The radio is a pre-certified Raytac MDBT50Q-P1MV2 module, not the nRF52840 die and not the PCA10056 development kit.
This is the circuit only. The module footprint and the antenna keep-out are still a layout job.

Power

File: power.kicad_sch

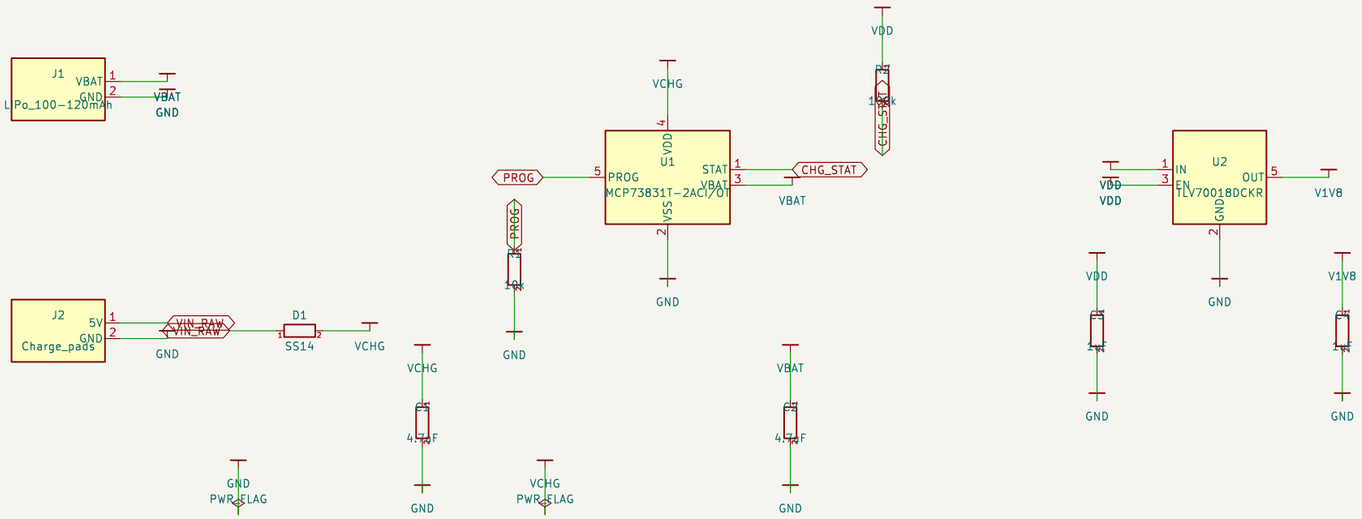
Radio

File: mcu.kicad_sch

Sensors

File: sensors.kicad_sch

SmartBand		
Sheet: /		
File: smartband.kicad_sch		
Title: SmartBand rev A		
Size: A3	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4



Rev A power.
J2 is the 5 V charge contact. D1 blocks a reversed charger.
arges a single LiPo to 4.20 V at 100 mA (R1 = 10k, I = 1000 V / Rprog).
the module VDDH pin. Do not tie VBAT to a 3.3 V–only rail: a full cell is 4.2 V.
ule REGO output. Set UICR.REGOUT0 to 3.3 V (value 5). The erased default is 1.8 V.
makes 1.8 V for the MAX30102 digital supply only. PPG LEDs use VBAT.
CHG_STAT is the charger open-drain status, pulled up to VDD.

SmartBand

Sheet: /Power/
File: power.kicad_sch

Title: Power, charge, and 1.8 V

Size: A3	Date: 2026-10-18	**Rev: A**
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VBAT
C5
10nF
GND
VBAT
C6
10nF
DCCH
L1
10uH
VDD
VDD
C7
10nF
GND
VDD
C8
10nF
GND
VDD
R5
100k
nRESET
SW1
RESET
nRESET
C9
100nF
GND
XL1
Y1
32.768kHz
XL2
XL1
C4
12pF
GND
XL2
C10
12pF
GND
VDD
C12
100nF
VDD
GND
VDD
VDDH
DCCH
U3
MDBT50Q-P1MV2
1 GND
2 GND
BTN GND
4 P1.10
OLED_RES 4 P1.11
15 GND
I2C_SDA GND 16 P0.27
XL1 17 P0.00/XL1
XL2 18 P0.01/XL2
I2C_SCL 19 P0.26
HAPTIC 20 P0.04
CHG_STAT 22 P0.06
FG_ALRT 25 P1.08
IMU_INT 27 P0.11
29 P0.12 PPG_INT
40 nRESET nRESET
41 P0.17 QSPI_CS
42 P0.19 QSPI_SCK
43 P0.21 QSPI_IO1
44 P0.20 QSPI_IO0
45 P0.23 QSPI_IO3
46 P0.22 QSPI_IO2
51 SWDIO SWDIO
53 SWDCLK SWDCLK
55 GND
GND
32 VBUS
33 GND
GND
QSPI_CS 1 CS
QSPI_IO1 2 SO/SIO1
QSPI_IO2 3 WP/SIO2
QSPI_IO0 5 SI/SIO0
QSPI_SCK 6 SCLK
QSPI_IO3 7 HOLD/SIO3
U4
MX25R6435FM2IL0
8 VCC
4 GND
GND
VDD
GND
SWDIO
nRESET
SWDCLK
1 VTref
2 SWDIO
3 nRESET
4 SWCLK
5 GND
J3
SWD_2030-NL
GND

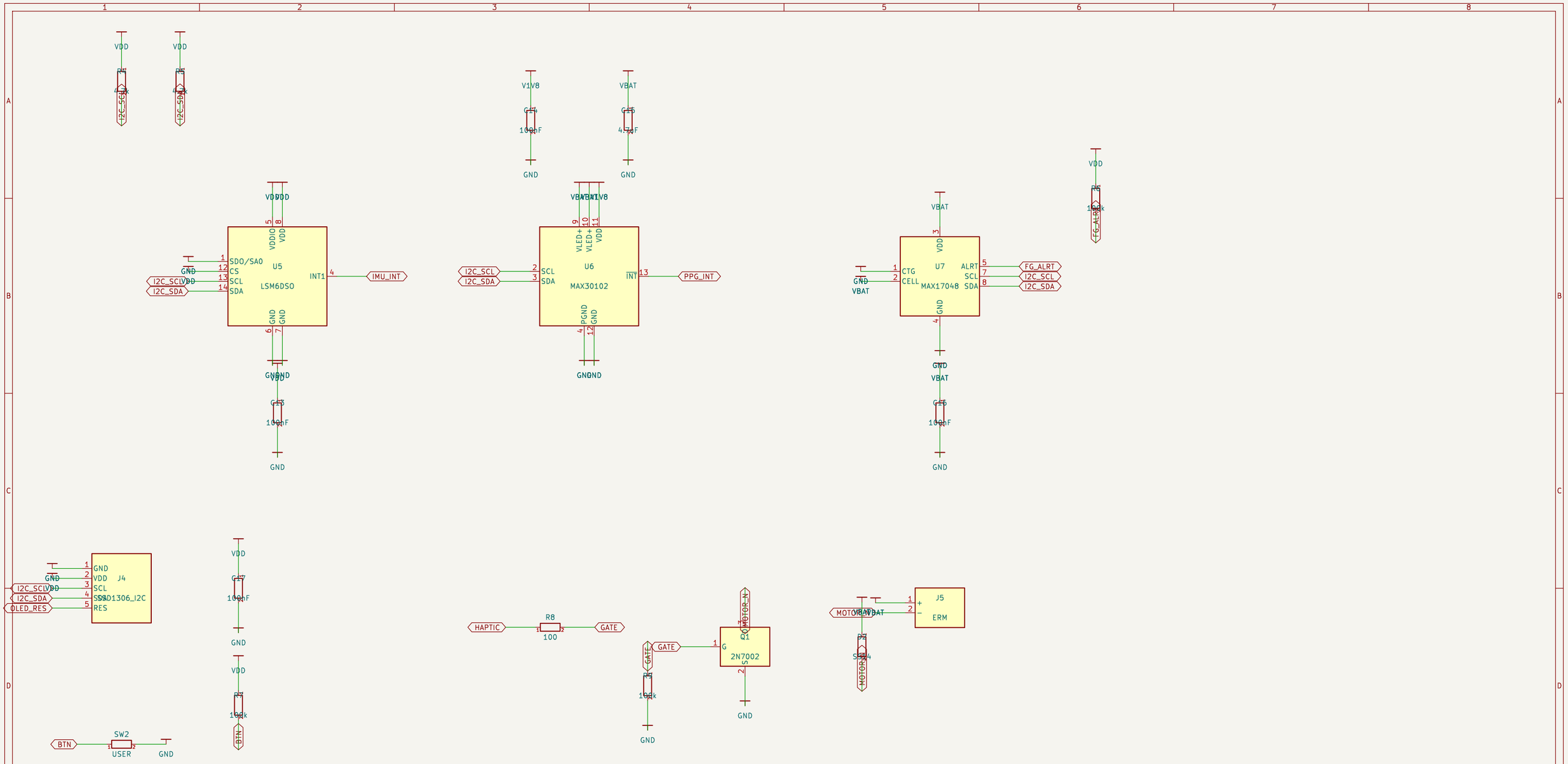
RF52840, PCB antenna). 32 MHz crystal and the radio DC/DC inductor are inside the module.
to VDDH, 10 uH from DCCH to VDD, 10 uF on VDDH and on VDD (Raytac section 8.1, L4 = 10 uH).
USB is unused. D+ and D− are open. VBUS is tied to GND.
pen. Keep copper off every layer under the antenna end and place that end at the board edge.
pins Raytac marks for QSPI: P0.19 SCK, P0.20 IO0, P0.21 IO1, P0.22 IO2, P0.23 IO3, P0.17 CS.
nRESET is P0.18 (module pin 40), so chip-select is P0.17 and not P0.18.
ytac says to delete the crystal and its load caps when the internal 32.768 kHz RC is used.
ned here are no-connect, including the low-frequency-only GPIOs that are not used.

SmartBand

Sheet: /Radio/
File: mcu.kicad_sch

Title: nRF52840 module, clock, debug, QSPI

Size: A3	Date: 2026-10-18	**Rev: A**
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One I2C bus, pulled up to VDD.
6A: SDO/SA0 tied to GND, CS tied to VDD, SPI pins SDx/SCx open. INT1 is the data-ready line.
upply is V1V8 (1.7 V to 2.0 V). LED anodes are VBAT. Stop PPG when the cell is under 3.2 V.
ll. ALRT is open-drain and is pulled up to VDD, not to VBAT. QSTRT is left open (internal pull-up).
ND, VDD, SCL, SDA, RES for an I2C SSD1306 panel. Confirm the panel pin order before layout.
utton is active-low. The coin motor returns through Q1; D2 is the flyback diode.

SmartBand	
Sheet: /Sensors/	
File: sensors.kicad_sch	
Title: IMU, PPG, fuel gauge, OLED, button, haptic	
Size: A3	Date: 2026-10-18
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: A**
	Id: 4/4